



AN2DL - First Homework Report NNMD

Nicolò Giallongo, Davide Goretti, Marco Fumagalli,

nickgiallongo, davidegoretti, marcofumagalli,

245279, 275499, 274158,

November 24, 2024

1. Introduction

This project tackles an *image classification problem* using **deep learning techniques**, with the goal of classifying blood cell images into eight categories. The key steps involved are:

- **Dataset Preparation:** Visualizing and cleaning the dataset to remove irrelevant or contaminating images.
- **Data Augmentation:** Applying data augmentation techniques to address class imbalance and expand the dataset.
- **Transfer Learning & Fine-Tuning:** Adapting and fine-tuning a pre-trained model (in our case EfficientNet) for the specific task of blood cell classification.

2. Problem Analysis

2.1 Dataset Characteristics

- Random images from the dataset were inspected to identify distinctive features and select appropriate augmentation techniques. Additionally, the class distribution was plotted, revealing a slight imbalance in the number of samples per class.
- Dataset cleaning addressed contamination caused by two irrelevant images, deemed unsuitable for training the neural network.

2.2 Validation

- The removal of unsuitable images caused minor changes in class distributions, with one label being disproportionately affected, introducing challenges in maintaining balance between training and validation sets.
- A **threshold-based splitting** was applied, dividing the dataset into a training set and a validation set. The splitting for each class was based on a threshold: if the percentage of the population of a class was greater than the threshold, a smaller portion of it was selected for the validation set. In this way, we slightly reduced the unbalance in the validation, while incremented it in the training set. This choice will be explained in the **Adaptive Data Augmentation** section. Subsequently the validation set was further divided into an effective validation set, an augmented test set and a standard test set. This division was designed to provide both a validation set and two test sets (one augmented and one standard) to analyze how the model's accuracy varies with data augmentation. This approach prepares for a potential real-world test set that might be particularly challenging to classify.

2.3 Adaptive Data Augmentation

- To address class imbalance, we employed an *adaptive augmentation strategy* during training, applying the same augmentations across all classes but varying the augmentation percentage. Underrepresented classes were augmented more extensively than overrepresented ones. This approach was fo-

cused on the training set to mitigate data imbalance introduced by the creation of the validation set. Prioritizing a higher proportion of original images in the validation and test sets, we accepted a greater reliance on augmented images in the training set.

- For the **augmented test set**, we applied a *uniform augmentation strategy*, where each class was augmented by the same percentage. This ensured the original class distribution remained intact while providing a larger and more diverse dataset for evaluation.

Table 1: Accuracy and loss.

	Test Set	Augmented Test Set
Accuracy	0.9152	0.1740
Loss	0.2593	2.2760

3. Model Building

3.1 Base Model Selection

- **EfficientNetB3** was chosen due to its efficiency and state-of-the-art performance in image classification tasks, including medical imaging.
- EfficientNet outperforms traditional models like VGG16 and ResNet50 by achieving higher accuracy with fewer parameters and lower computational cost.
- The model's *scaling method* (width, depth, and resolution scaling) ensures optimal use of resources, improving classification performance for complex datasets (like blood cells images).
- EfficientNet has shown superior results in various domains, making it a robust choice for image classification tasks with limited data.
- Pre-trained weights on ImageNet were used to leverage general feature extraction capabilities, allowing the model to adapt efficiently to the blood cell classification task.

3.2 Model Architecture

We designed the architecture of our model to efficiently extract and process features from input images while minimizing the risk of overfitting. The following components were integrated to ensure optimal performance and generalization:

- **Input Layer:** The model takes as input RGB images of size 96x96, representing the dataset's dimensions.

- **Base Model (EfficientNetB3):** We used a pre-trained EfficientNetB3 model, excluding its top layer, as the feature extractor. This model was selected for its high performance-to-parameter ratio. To retain the pre-learned knowledge and accelerate convergence, we froze its weights during training, thus avoiding overfitting.
- **Global Average Pooling (GAP):** After the base model, we applied a GAP layer to reduce the spatial dimensions of the feature maps into a compact vector. This approach efficiently captures global image features while keeping the parameter count low, reducing the risk of overfitting compared to flattening.
- **First Dense Layer (512 Neurons):** A fully connected layer with 512 neurons was added to learn complex patterns from the extracted features. This size balances model capacity and computational efficiency, ensuring robust feature learning.
- **Batch Normalization and ReLU:** To stabilize training and accelerate convergence, we applied batch normalization after the dense layer. The ReLU activation function was used to introduce non-linearity and prevent the vanishing gradient problem.
- **Dropout:** A dropout layer was incorporated after the dense layer to randomly deactivate neurons during training, reducing overfitting and encouraging the model to learn generalized patterns.
- **Second Dense Layer (256 Neurons):** We included a second dense layer with 256 neurons to refine the features further, enhancing the model's ability to learn abstract representations for classification.
- **Second Batch Normalization and ReLU:** Batch normalization and ReLU activation were applied again after the second dense layer to maintain stability and non-linearity.
- **Second Dropout Layer:** Another dropout layer was added to provide additional regularization, particularly as the model's complexity increased.
- **Output Layer (Softmax):** The final layer consists of 8 neurons corresponding to the 8 blood cell classes. A softmax activation function was used to output normalized probabilities for each class.

4. Method

4.1 Fine-Tuning and Regularization

To mitigate overfitting and enhance the model’s performance, the strategy was adjusted as follows:

- At the start of training, we allowed the model to train freely without imposing major constraints. This approach enabled the network to establish initial neural connections and learn fundamental patterns in the data. We experimented with different optimizers, including Adam and RMSProp, to assess their impact on the training process. After comparison, we found that the SGD optimizer provided the best results in terms of convergence speed and overall performance.
- As training progressed, we carefully monitored the accuracy and loss curves for both the training and validation sets. Based on these observations, we dynamically adjusted the dropout rate to optimize regularization, balancing between preventing overfitting and allowing sufficient learning capacity.
- Along with adjusting the dropout rate, we also changed the learning rate and momentum parameters to accelerate convergence while maintaining stability. These iterative adjustments allowed the model to adapt efficiently during the critical stages of training.
- We used the Keras callback `ReduceLROnPlateau` to dynamically lower the learning rate when validation performance plateaued. This adjustment aimed to improve convergence by allowing the model to fine-tune its parameters more efficiently during the later stages of training.

- When validation accuracy plateaued or improved only slightly, we increased data augmentation for both the training and validation sets. Care was taken to ensure that augmentation did not reduce the accuracy on the non-augmented test set, maintaining a balance between augmented and non-augmented test performance.
- Eventually, we gradually unfroze the EfficientNet layers to allow fine-tuning and repeated the process from the beginning.

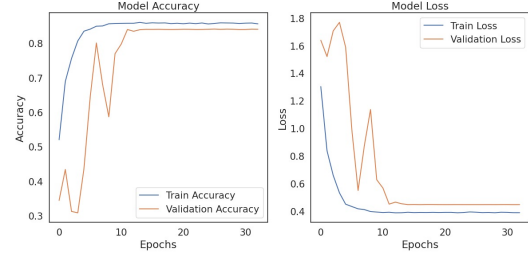


Figure 1: Accuracy and loss on validation and training sets.

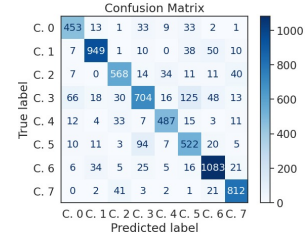


Figure 2: Confusion matrix.

5. Conclusion

This project successfully addressed the challenge of classifying blood cell images into eight categories using deep learning techniques. Key steps included dataset preparation, adaptive data augmentation to mitigate class imbalance, and leveraging transfer learning with the EfficientNetB3 model.

EfficientNetB3 proved to be an optimal choice for this task due to its efficiency and high performance in image classification, particularly with medical imaging datasets. By employing adaptive data augmentation and fine-tuning, the model achieved robust performance while addressing overfitting concerns.

The results highlight the importance of balancing data preprocessing, model architecture, and regularization techniques in achieving accurate and generalizable classifications. This approach demonstrates a practical workflow for similar image classification challenges in medical and other domains.